

DehatiAI

Multilingual Agricultural Intelligence & Market Infrastructure for Punjab

An offline-resilient, voice-enabled AI platform translating agronomic science and real-time Mandi intelligence into spoken Urdu and Punjabi for smallholder empowerment across 36 districts.

12.5M

Target Smallholders

Empowering smallholder farmers across Punjab's rural agrarian economy.

90%

Cost Reduction via dual-tier PostgreSQL query caching & Anthropic prompt caching.

Urdu + Punjabi

Voice-First UI

Zero-literacy accessibility with Web Speech API audio synthesis.

Systemic Bottlenecks in Punjab Agriculture

24.0%

GDP Contribution

Agriculture employs 37.4% of Pakistan's labor force, yet extension services remain severely under-resourced (Economic Survey 2023-24).

35% - 40%

Pre-Harvest Yield Loss

Smallholders suffer massive losses annually due to delayed disease identification and unscientific chemical dosage (UN FAO).

46%

Connectivity Gap

46% of rural areas lack continuous 4G coverage. Cloud-only apps fail when internet drops in remote fields (PTA 2024).

Key Structural Bottlenecks Solved by DehatiAI

Language Barrier

Over 50% rural illiteracy. Existing apps rely on text-heavy English menus instead of voice.

Mandi Asymmetry

Information gaps with commission agents (Arhtis) cost farmers 15-20% farm-gate margin.

High Extension Ratio

Extension Officer ratio to farmers is ~1:1,500, preventing timely on-field physical visits.

Codebase-Verified Technical Stack

Client Layer (Vite React PWA)

- **Service Worker PWA Shell:** Offline caching for HTML/CSS/JS assets
- **Native Speech API:** Voice recognition in Urdu (ur-PK) & Punjabi (pa-PK)
- **Noto Nastaliq Urdu Typography:** High-readability regional script rendering

API Router Layer (Node.js Express)

- **Rate Limiter:** 30 AI calls/hr per IP to protect token expenditure
- **Agriculture Keyword Guard:** Rejects non-farming prompts instantly
- **SSE Stream Heartbeat:** Real-time streaming response cleanup

Intelligence Layer (Claude 3.5 Sonnet Vision)

- **Model:** c1aude-3-5-sonnet-20241022 (#1 Vision Model)
- **Ephemeral Prompt Cache:** 5-minute system prompt caching (90% savings)
- **Diagnostic Protocol:** Inspects leaf lesions & prescribes registered sprays

Data & Resilience Layer (Postgres + Supabase)

- **Primary DB:** PostgreSQL pool on Railway with L2 query cache
- **Fallback DB:** Supabase REST API secondary fallback
- **Audit Log:** Immutable admin logs (admin_audit_logs table)

Exchange Rate: 1 USD = 278 PKR

Infrastructure Operating Cost Breakdown (PKR)

SERVICE	PURPOSE	PRICING MODEL	FREE LIMIT	CURRENT (MO)	500 DAU (MO)	5,000 DAU (MO)	5,000 DAU (YR)
Anthropic Claude API	AI Chat, Vision, Disease	Per-Token Usage	Rs. 0 (Pay-as-you-go)	Rs. 695	Rs. 43,785	Rs. 312,750	Rs. 3,753,000
Railway App Server	Express API Server Engine	Compute (CPU/RAM)	Rs. 1,390 Trial Credit	Rs. 0	Rs. 3,336	Rs. 12,510	Rs. 150,120
Railway PostgreSQL	Primary DB & L2 Cache	Storage & IOPS	Included in Railway	Rs. 0	Included	Included	Included
Vercel PWA Hosting	Frontend CDN & PWA	Bandwidth & Builds	100 GB / month	Rs. 0	Rs. 0	Rs. 5,560	Rs. 66,720
Supabase Auth/Backup	Secondary Fallback Storage	50k MAU Free Tier	Rs. 0	Rs. 0	Rs. 0	Rs. 0	
Open-Meteo Weather	Weather Forecast Feed	Free Public API	10,000 calls / day	Rs. 0	Rs. 0	Rs. 0	Rs. 0
AMIS Mandi Feed	Commodity Market Rates	Internal DB Storage	Unlimited	Rs. 0	Rs. 0	Rs. 0	Rs. 0
Custom .PK Domain	Branded Domain & SSL	Flat Annual Renewal	N/A	Rs. 0	Rs. 278	Rs. 280	Rs. 3,360
TOTAL BUDGET	Complete Infrastructure	Combined	—	Rs. 695	Rs. 47,400	Rs. 331,100	Rs. 3,973,200